

16.4

→ Type 1:-

$$\int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$$

$$\rightarrow \cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\rightarrow \sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right)$$

$$\rightarrow C : |z| = 1, \quad z = re^{i\theta}$$

$$\rightarrow d\theta = \frac{dz}{iz}$$

$$\oint_C f(z) \frac{dz}{iz} \quad \stackrel{2\pi}{\circ} \int_0^{2\pi} \frac{\cos 3x}{z^2 + 1} dx$$

Step 1:-

→ Simply $f(z)$

$$\int_0^{2\pi} \frac{1}{\sqrt{2} - \cos \theta} d\theta$$

Step 2:-

→ find singularities lying in $C : |z| = 1$

Step 3:-

find residues at singularities

Step 4:-

Use Residue Theorem

$$\oint_C f(z) dz = 2\pi i \sum_{j=1}^K \text{Res}(f(z))$$



Type 2:-

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res}(f(z))$$

1) Step 1:-

Consider $f(z)$ replace x by z

2) Step 2:-

Singularities in upper half plane

3) Step 3:-

Residues at singularities in upper half plane.

o) Step 4:-

Use $\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res}(f(z))$



When: $\int_0^{\infty} f(x) dx$

Step 1:-

Consider $\int_{-\infty}^{\infty} f(x) dx$

$$\therefore \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

Step 2:-

Consider $f(z)$

Step 3:-

Singularities in upper half plane

Step 4:-

Residue at singularities in upper half plane

Step 5:-

$$\int_{-\infty}^{\infty} f(x) dx = \left(2\pi i \sum \text{Res } f(z) \right) \times \frac{1}{2}$$

Fourier integrals

Steps:-

$$\int_{-\infty}^{\infty} \cos sx \, f(x) \, dx, \quad \int_{-\infty}^{\infty} \sin sx \, f(x) \, dx$$

Step 1:-

Consider

$$\Rightarrow f(x) \times e^{isx}$$

s is any real no

Step 2:-

Consider

$$f(z) \times e^{isz}$$

Step 3:-

Singularities in upper half Plane

Step 4:-

Residues at singularities in upper half Plane

$$\text{Res}_{z=z_0} (f(z) e^{isz})$$

Step 5:-

$$\int_{-\infty}^{\infty} f(x) e^{isx} \, dx = 2\pi i \sum \text{Res}(f(z) e^{isz})$$

16.4

→ Residue integration of real integrals:-

Surprisingly, residue integration can also be used to evaluate certain classes of complicated real integrals. This shows an advantage of complex analysis over real analysis or calculus.

Integrals of Rational function of $\cos \theta$ and $\sin \theta$:-

Type 1 :-

$$J = \int_0^{2\pi} F(\cos \theta, \sin \theta) d\theta$$

where $F(\cos \theta, \sin \theta)$ is a real function of $\cos \theta$ and $\sin \theta$

(e.g. $\frac{\sin^2 \theta}{5 - 4 \cos \theta}$) and is finite

(does not become infinite) on the

interval of integration. Setting $z = e^{i\theta}$
we obtain,

$$\cos \theta = \frac{1}{2} (e^{i\theta} + e^{-i\theta})$$

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

Similarly;

$$\sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$$

$$= \frac{1}{2i} \left(\frac{1}{z} - \frac{1}{z} \right)$$

Since "F" is rational in $\cos \theta$
and $\sin \theta$

Above relation of $\cos \theta$ and $\sin \theta$ shows that F is now a rational function of z
say $f(z)$ as

$$z = e^{i\theta}$$

So,

$$\rightarrow dz = ie^{i\theta} d\theta, \quad d\theta = \frac{dz}{ie^{i\theta}} = \frac{dz}{iz}$$

$$\Rightarrow \int_0^{2\pi} F(\cos\theta, \sin\theta) d\theta \quad \text{takes}$$

the form,

$$J = \oint_C f(z) \frac{dz}{iz}$$

as θ ranges from "0 to 2π "

in $\int_0^{2\pi} F(\cos\theta, \sin\theta) d\theta$, the

variable $z = e^{i\theta}$ ranges counter-clockwise, once around the unit circle $|z| = 1$

$$C: |z| = 1$$

$$\circ \circ \rightarrow \oint_C f(z) \frac{dz}{iz}$$

$$\rightarrow \cos\theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$1) \operatorname{Im} Q = \frac{1}{2i} \left(z - \frac{1}{z} \right)$$

$$2) C: |z| = 1$$

Example 1:-

Integral of Type :-

$$\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos\theta} = 2\pi$$

L.H.S R.H.S

as

$$\rightarrow \boxed{\cos\theta = \left(\frac{1}{z} + z \right) \frac{1}{2}}$$

where

$$z = e^{i\theta}$$

$$dz = ie^{i\theta} d\theta$$

$$d\theta = \frac{dz}{ie^{i\theta}}$$

$$\because e^{i\theta} = z$$

$$\rightarrow \boxed{d\theta = \frac{dz}{iz}}$$

So;

$$\boxed{\oint_C f(z) \frac{dz}{iz}}$$

where $f(z) = \frac{1}{\sqrt{2} - \cos \theta}$

So,

$$\oint_C \frac{1}{\sqrt{2} - \cos \theta} \frac{dz}{iz}$$

and

$$\cos \theta = \frac{1}{2} \left[z + \frac{1}{z} \right]$$

and

$$C: |z| = 1$$

$$\oint_C \frac{1}{\left[\sqrt{2} - \frac{1}{2} \left(z + \frac{1}{z} \right) \right]} \frac{dz}{iz}$$

$$\oint_C \frac{1}{iz\sqrt{2} - \left(z^2 i + i \right) \frac{1}{2}} dz$$

$$\oint_C \frac{1}{iz\sqrt{2} - \frac{z^2 i}{2} - \frac{i}{2}} dz$$

$$\oint_C \frac{1}{\frac{i}{2} (z\sqrt{2} - z^2 - 1)} dz$$

$$= \oint_C \frac{1}{-\frac{i}{2} (z^2 - 2\sqrt{2}z + 1)} dz$$

$$= -\frac{2}{i} \oint_C \frac{1}{z^2 - 2\sqrt{2}z + 1} dz \rightarrow \textcircled{1}$$

finding singularities:-

$$z^2 - 2\sqrt{2}z + 1$$

$$a=1, b=-2\sqrt{2}, c=1$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$z = \frac{+2\sqrt{2} \pm \sqrt{(-2\sqrt{2})^2 - 4(1)(1)}}{2 \times 1}$$

$$z = \frac{2\sqrt{2} \pm \sqrt{4}}{2}$$

$$z = \frac{2(\sqrt{2} \pm 1)}{2}$$

$$z = \sqrt{2} \pm 1$$

So,

$$z = \sqrt{2} + 1 \quad \text{and} \quad z = \sqrt{2} - 1$$

are the singularities, the type of singularity is simple pole.

But

$\rightarrow z = \sqrt{2} + 1$ lies outside the given contour $C: |z| = 1$

and

$z = \sqrt{2} - 1$ lies inside the given contour $C: |z| = 1$

Equation (1) becomes;

$$= -\frac{2}{i} \oint_C \frac{1}{z^2 - 2\sqrt{2}z + 1} dz$$

where $z = \sqrt{2} \pm 1$ ✓

$$z - \sqrt{2} - 1 = 0$$

and ✓

$$z - \sqrt{2} + 1 = 0$$

$$= -\frac{2}{i} \oint \frac{1}{(z - \sqrt{2} - 1)(z - \sqrt{2} + 1)} dz$$

$$\operatorname{Re} (f(z)) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \left[\frac{d^{m-1}}{dz^{m-1}} \times ((z-z_0)f(z)) \right]$$

(lies inside
Contour)

↳ singularity that
lies inside contour

$$\operatorname{Re} (f(z)) = \frac{1}{(1-1)!} \lim_{z \rightarrow \sqrt{2}-1} \left[\frac{d^0}{dz^0} \times \frac{1}{(z-\sqrt{2}-1)(z-\sqrt{2}+1)} \right]$$

$$\operatorname{Re} (f(z)) = \lim_{z \rightarrow \sqrt{2}-1} \left[(z-\sqrt{2}+1) \times \frac{1}{(z-\sqrt{2}-1)(z-\sqrt{2}+1)} \right]$$

$$\operatorname{Re} (f(z)) = \lim_{z \rightarrow \sqrt{2}-1} \left[\frac{1}{z-\sqrt{2}-1} \right]$$

$$= \left(\frac{1}{\sqrt{2}-1-\sqrt{2}-1} \right)$$

$$= \frac{1}{-2}$$

$$\text{as } -2\phi \quad \oint \frac{1}{(z-\sqrt{2}-1)(z-\sqrt{2}+1)} dz$$

$$\Rightarrow = \frac{-2}{i} \times \frac{1}{-2}$$

$$= \frac{1}{i}$$

Using Residue theorem;

$$\oint_C f(z) dz = 2\pi i \sum_{j=1}^K \text{Res}(f(z))$$

$$= 2\pi i \times \left(\frac{-1}{2}\right) \times \left(\frac{-2}{i}\right)$$

$$= 2\pi$$

$$Q_3) \int_0^{2\pi} \frac{1 + \sin \theta}{3 + \cos \theta} d\theta$$

$$\rightarrow \sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right)$$

$$\rightarrow \cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\rightarrow d\theta = \frac{dz}{iz}$$

$$\rightarrow C : |z| = 1$$

$$\oint_C f(z) \frac{dz}{iz}$$

where, $f(z) = \frac{1 + \sin \theta}{3 + \cos \theta}$

$$f(z) = \frac{1 + \left(\frac{1}{2i} \right) \left(z - \frac{1}{z} \right)}{3 + \frac{1}{2} \left(z + \frac{1}{z} \right)}$$

So,

$$\oint_C \frac{1 + \left(\frac{1}{2i} \right) \left(z - \frac{1}{z} \right)}{3 + \frac{1}{2} \left(z + \frac{1}{z} \right)} \frac{dz}{iz}$$

$$= \oint_C \frac{1 + \frac{1}{2i} \left(\frac{z^2-1}{z} \right) dz}{3 + \frac{1}{2} \left(\frac{z^2+1}{z} \right) i z}$$

$$= \oint_C \frac{1 + \frac{z^2-1}{2iz} dz}{3 + \frac{z^2+1}{2z} iz}$$

$$= \oint_C \frac{2iz + z^2 - 1}{6z + z^2 + 1} \frac{dz}{iz}$$

$$= \oint_C \frac{2iz + z^2 - 1}{z(6z + z^2 + 1)} dz$$

$$= \oint_C \frac{2iz + z^2 - 1}{z(6z + z^2 + 1)} dz$$

Singularities :-

$z=0$ (Simple Pole)

$$z^2 + 6z + 1 = 0$$

where

$$a=1, b=6, c=1$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2 \times a}$$

$$z = \frac{-6 \pm \sqrt{36 - 4}}{2}$$

$$z = \frac{-6 \pm 4\sqrt{2}}{2}$$

$$z = \frac{2(-3 \pm 2\sqrt{2})}{2}$$

$$z = -3 \pm 2\sqrt{2}$$

$$z = -3 + 2\sqrt{2}, \quad z = -3 - 2\sqrt{2}$$

(Simple Poles)

But;

$$z = -3 + 2\sqrt{2} \text{ (lies inside } C)$$

where

$$z = -3 - 2\sqrt{2} \text{ (lies outside } C)$$

We have two singularities (simple Poles)
 $z=0$, $z=-3+2\sqrt{2}$ that lies inside given contour.

So,
$$\oint_C \frac{z^2 + 2iz - 1}{z(z+3+2\sqrt{2})(z+3-2\sqrt{2})} dz$$

Finding residues :- at simple $[z=0]$

$$\text{Res}_{z=0}(f(z))$$

$$\text{Res}_{z=0}(f(z)) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \left[\frac{d^{m-1}}{dz^{m-1}} ((z-z_0)(f(z))) \right]$$

$$= \frac{1}{(1-1)!} \lim_{z \rightarrow 0} \left[\frac{d^0}{dz^0} (z-0) \times \frac{z^2 + 2iz - 1}{z(z+3+2\sqrt{2})(z+3-2\sqrt{2})} \right]$$

$$= \lim_{z \rightarrow 0} \left[z \times \frac{z^2 + 2iz - 1}{z(z+3+2\sqrt{2})(z+3-2\sqrt{2})} \right]$$

$$= \frac{0 + 2 \times i \times 0 - 1}{(0 + 3 + 2\sqrt{2})(3 - 2\sqrt{2} + 0)}$$

$$= -1$$

$$\left. \begin{aligned} \text{Res}_{z=z_0} f(z) \\ = \frac{p(z_0)}{q'(z_0)} \end{aligned} \right\}$$

Similarity Residue at simple

Pole $z = -3 + 2\sqrt{2}$

as $f(z) = \frac{z^2 + 2iz - 1}{z^3 + 6z^2 + z}$ is

in P/Q form so;

$$\text{Res}(f(z))_{z = -3 + 2\sqrt{2}} = \frac{P(z_0)}{Q'(z_0)}$$

$$\text{Res}(f(z))_{z = -3 + 2\sqrt{2}} = \frac{(z^2 + 2iz - 1)|_{z = -3 + 2\sqrt{2}}}{(3z^2 + 12z + 1)|_{z = -3 + 2\sqrt{2}}}$$

$$\begin{aligned}\text{Res}(f(z))_{z = -3 + 2\sqrt{2}} &= \frac{(-3 + 2\sqrt{2})^2 + 2(-3 + 2\sqrt{2})i - 1}{3(-3 + 2\sqrt{2})^2 + 12(-3 + 2\sqrt{2}) + 1} \\ &= 1 + 0.35i \checkmark\end{aligned}$$

Applying Residue theorem:-

$$\begin{aligned}\oint_C f(z) dz &= 2\pi i \times \sum_{j=1}^K \text{Res}(f(z)) \\ &= 2\pi i \times (-1 + 1 + 0.35i) \\ &= -2.199\end{aligned}$$

Type 2:-

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res } f(z)$$

→ (When the singularity lies in the upper half plane) ←

Example :-

$$\int_0^{\infty} \frac{1}{1+x^4} dx$$

→ If $f(x)$ is even function

then;

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

$$\neq \int_{-\infty}^{\infty} \frac{1}{1+x^4} dx$$

$$= 2\pi i \sum \text{Res} \left(\frac{1}{1+z^4} \right)$$

finding Poles:-

$$\rightarrow \frac{1}{1+z^4}$$

$$\rightarrow \frac{1}{1+z^4}$$

$$1+z^4=0$$

$$z^4 = -1$$

$$z = (-1)^{1/4} \quad n=1,$$

finding the roots

$$w = R(\cos \phi + i \sin \phi)^n$$

$$R = r^{1/n}$$

$$R = r^{1/4}$$

$$R = (1)^{1/4} = 1$$

where;

$$\phi = \frac{\pi + 2K\pi}{n}$$

$$K=0,1,2,3$$

$$K=0$$

$$\phi = \frac{\pi + 2 \times 0 \pi}{4}$$

$$\rightarrow \phi_1 = \frac{\pi}{4}$$

when

$$|K=1|$$

$$\rightarrow \phi_2 = \frac{\pi + 2 \times 1 \pi}{4}$$

$$\phi_2 = \frac{3\pi}{4}$$

$$|K=2|$$

$$\rightarrow \phi_3 = \frac{\pi + 2 \times 2 \pi}{4} = \frac{5\pi}{4}$$

$$|K=3|$$

$$\rightarrow \phi_4 = \frac{\pi + 2 \times 3 \pi}{4} = \frac{7\pi}{4}$$

where

$$\bullet) W_1 = 1 \times \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right) = e^{i\pi/4}$$

$$\bullet) W_2 = 1 \times \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right) = e^{i3\pi/4}$$

$$\bullet) W_3 = 1 \times \left(\cos\left(\frac{5\pi}{4}\right) + i \sin\left(\frac{5\pi}{4}\right) \right) = e^{i5\pi/4}$$

$$\Rightarrow w_4 = 1 \times \left(\cos\left(\frac{7\pi}{4}\right) + i \sin\left(\frac{7\pi}{4}\right) \right) = e^{i7\pi/4}$$

→ Only $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ in upper half

Plane. So, only w_1 and w_2 poles are required. The w_1 and w_2 are simple poles.

find $\text{Res}(f(z))$ at simple Poles:-

$$\Rightarrow \text{Res}_{z=e^{i\pi/4}} \left(\frac{1}{1+z^4} \right) = \frac{P(z_0)}{Q'(z_0)} = \frac{1}{4z^3}$$

$$\text{Res}_{z=e^{i\pi/4}} \left(\frac{1}{1+z^4} \right) = \frac{1}{4(e^{i\pi/4})^3}$$

$$\text{Res}_{z=e^{i\pi/4}} \left(\frac{1}{1+z^4} \right) = \frac{1}{4 e^{i3\pi/4}}$$

$$\text{Res}_{z=e^{i\pi/4}} \left(\frac{1}{1+z^4} \right) = \frac{1}{4} e^{-i\frac{3\pi}{4}}$$

$$\therefore \text{At } w = e^{i \frac{3\pi}{4}}$$

$$\text{Res}_{z = e^{i 3\pi/4}} \left(\frac{1}{1+z^4} \right) = \frac{p(z_0)}{Q'(z_0)} = \frac{1}{4z^3}$$

$$\text{Res}_{z = e^{i 3\pi/4}} \left(\frac{1}{1+z^4} \right) = \frac{1}{4 \times (e^{i 3\pi/4})^3}$$

$$\begin{aligned} \text{Res}_{z = e^{i 3\pi/4}} \left(\frac{1}{1+z^4} \right) &= \frac{1}{4 \times e^{i 9\pi/4}} \\ &= \frac{1}{4} e^{-i \frac{9\pi}{4}} \end{aligned}$$

Simplifying :-

$$\therefore \frac{1}{4} e^{-i \frac{9\pi}{4}}$$

$$= \frac{1}{4} e^{\frac{-8\pi i - \pi i}{4}}$$

$$= \frac{1}{4} \left(e^{-2\pi i} \cdot e^{-\pi i/4} \right)$$

$$= \frac{1}{4} \left(1 \times e^{-\pi i/4} \right)$$

$$\cdot) \frac{1}{4} e^{-\frac{3i\pi}{4}}$$

$$= \frac{1}{4} e^{-\frac{4i\pi + i\pi}{4}}$$

$$= \frac{1}{4} \left[e^{-\frac{i\pi}{4}} \cdot e^{i\pi/4} \right]$$

$$= \frac{1}{4} \left[-1 \times e^{i\pi/4} \right]$$

$$= -\frac{1}{4} e^{i\pi/4}$$

So, ∞

$$\int_{-\infty}^{\infty} \frac{1}{1+x^4} dx = 2\pi i \sum \text{Res } f(z)$$

$$= 2\pi i \left[+\frac{1}{4} e^{-\frac{\pi i}{4}} - \frac{1}{4} e^{\frac{\pi i}{4}} \right]$$

$$= -\frac{1}{4} 2\pi i \left(-e^{-\frac{\pi i}{4}} + e^{i\pi/4} \right)$$

$$= \frac{-2 \times 2\pi i}{4} \left[-e^{\frac{-i\pi}{4}} + e^{\frac{i\pi}{4}} \right]$$

$$= \frac{-2 \times 2\pi i \times i}{4} \left[\frac{e^{\frac{-i\pi}{4}} + e^{\frac{i\pi}{4}}}{2i} \right]$$

$$= \frac{2\pi^2}{4} \left[\sin\left(\frac{\pi}{4}\right) \right]$$

$$= \frac{\pi}{2} \times \frac{\sqrt{2}}{2}$$

$$= \frac{+\pi}{\sqrt{2}}$$

important step

$$= \frac{1}{2} \times \frac{\pi}{\sqrt{2}}$$

$$= \frac{\pi}{2\sqrt{2}}$$

Nov 1 / 22

Fourier integrals:-

Previous lecture Review:-

Type 1:-

$$\int_0^{2\pi} F(\sin \theta, \cos \theta) d\theta$$

where,

$$\rightarrow \cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$$

$$\rightarrow \sin \theta = \frac{1}{2} \left(z - \frac{1}{z} \right)$$

$$\rightarrow C: |z| = 1$$

$$\rightarrow z = e^{i\theta}$$

$$\rightarrow d\theta = \frac{dz}{iz}$$

$$\rightarrow \oint_C f(z) \frac{dz}{iz}$$

Type 2:-

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res } f(z)$$

↳ Singularities lies in the upper half plane.

→ Fourier integrals :-

$$1) \int_{-\infty}^{\infty} f(x) \cos sx dx$$

$$2) \int_{-\infty}^{\infty} f(x) \sin sx dx$$

→ (where s is real) ←

from type 2:-

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res } f(z)$$

So,

$$\int_{-\infty}^{+\infty} f(x) e^{isx} dx = 2\pi i \sum \text{Res} (f(z) e^{isz})$$

where, we sum the residues of $f(z)e^{isz}$ at its poles in the upper half plane. Equating, the real and imaginary parts on both sides of above equation

$$\rightarrow \int_{-\infty}^{\infty} f(x) \cos sx \, dx = -2\pi \sum \text{Im Res} [f(z)e^{isz}]$$

$$\rightarrow \int_{-\infty}^{\infty} f(x) \sin sx \, dx = +2\pi \sum \text{Re Res} [f(z)e^{isz}]$$

where;

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) [\sin \cos sx + i \sin sx] \, dx \\ = 2\pi i \sum \text{Res} (f(z)e^{isz}) \end{aligned}$$

Example:-

$$\int_{-\infty}^{+\infty} \frac{\cos x}{K^2 + x^2} dx = \frac{\pi}{K} e^{-Ks}$$

$$\int_{-\infty}^{\infty} \frac{\sin x}{K^2 + x^2} dx = 0$$

($s > 0, K > 0$)

Consider

$$\frac{1}{K^2 + x^2} e^{isx} \Rightarrow \frac{1}{K^2 + z^2} e^{isz}$$

Singularity

$$K^2 + z^2 = 0$$

$$z^2 = -K^2$$

$$z = \pm Ki$$

The only pole (Simple Pole) which lies in upper half plane is

$$z = Ki \text{ (Simple Pole)}$$

finding Residue at $z = Ki$:-

$$\text{Res}_{z=Ki} \left(\frac{e^{isz}}{K^2 + z^2} \right) = \frac{P(z_0)}{Q'(z_0)}$$

$$= \frac{e^{isz(z_0)}}{2(z_0)}$$

$$= \frac{e^{is(Ki)}}{2(Ki)}$$

$$= \frac{e^{-Ks}}{2Ki} \times \frac{i}{i}$$

$$= - \frac{e^{-Ks}}{2K} i$$

$$\text{as } +\infty \int_{-\infty}^{\infty} f(x) e^{isz} dx = 2\pi i \sum \text{Res}(f(z) e^{isz})$$

$$\int_{-\infty}^{\infty} \frac{e^{isx}}{x^2 + K^2} dx = 2\pi i \left(- \frac{e^{-Ks}}{2K} i \right)$$

$$\int_{-\infty}^{\infty} \frac{e^{isx}}{K^2 + x^2} dx = \frac{\pi}{K} e^{-Ks}$$

$$\int_{-\infty}^{\infty} \frac{\cos sx + i \sin sx}{K^2 + x^2} dx = \frac{\pi}{K} e^{-Ks}$$

Comparing :-

$$\int_{-\infty}^{\infty} \frac{\cos sx}{K^2 + x^2} dx = \frac{\pi}{K} e^{-Ks}$$

and

$$\int_{-\infty}^{\infty} \frac{\sin sx}{K^2 + x^2} dx = 0$$

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Q10:-
$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^3} dx$$

$$f(z) = \frac{1}{(1+z^2)^3}$$

$$f(z) = \frac{1}{(1+z^2)^3}$$

Singularities :-

$\rightarrow z = \pm i$ Pole of order 3

The $z = i$ (lies in the upper half plane)

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res}(f(z))$$

finding $\text{Res}(f(z))$ at $z = i$:-

$$\text{Res}_{z=i} \left(\frac{1}{(1+z^2)^3} \right)$$

$$\text{Res}_{z=i} \left(\frac{1}{(1+z^2)^3} \right) = \frac{1}{(m-1)!} \lim_{z \rightarrow i} \left[\frac{d^{m-1}}{dz^{m-1}} \left((z-i)^3 f(z) \right) \right]$$

$$= \frac{1}{(3-1)!} \lim_{z \rightarrow i} \left[\frac{d^{3-1}}{dz^{3-1}} \left((z-i)^3 \times \frac{1}{(1+z^2)^3} \right) \right]$$

$$= \frac{1}{2!} \lim_{z \rightarrow i} \left[\frac{d^2}{dz^2} \left(\frac{1}{(z+i)^3 (z-i)^3} \right) \right]$$

$$= \frac{1}{2} \lim_{z \rightarrow i} \left[\frac{d^2}{dz^2} \frac{1}{(z+i)^3} \right]$$

$$= \frac{1}{2} \lim_{z \rightarrow i} \left[\frac{d}{dz} \frac{-3}{(z+i)^4} \right]$$

$$= \frac{1}{2} \lim_{z \rightarrow i} \left[\frac{-3 \times -4}{(z+i)^5} \right]$$

$$= \frac{12}{2} \lim_{z \rightarrow i} \left[\frac{1}{(z+i)^5} \right]$$

$$= 6 \left(\frac{1}{(1+i)^5} \right)$$

$$= 6 \left(\frac{1}{(2i)^5} \right)$$

$$= \frac{6}{32i^5}$$

$$= \frac{6}{32i}$$

$$= \frac{3}{16i}$$

Now,

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^3} dx = 2\pi i \times \frac{3}{16i}$$

$$= \frac{3\pi}{8}$$